

Foundations of K-nearest neighbors algorithm

[K-nearest neighbors algorithm]

1.0 Content Block

A commonly used distance metric for continuous variables is Euclidean distance. For discrete variables, such as for text classification, another metric can be used, such as the overlap metric (or Hamming distance). In the context of gene expression microarray data, for example, k-NN has been employed with correlation coefficients, such as Pearson and Spearman, as a metric. Often, the classification accuracy of k-NN can be improved significantly if the distance metric is learned with specialized algorithms such as large margin nearest neighbor or neighborhood components analysis.

2.0 Content Block

for constants B_1 and B_2 where $s_n^2 = \sum_{i=1}^n w_{ni}^2$ and $t_n = n - 2/d \sum_{i=1}^n w_{ni} \{i^{1+2/d} - (i-1)^{1+2/d}\}$. The optimal weighting scheme $\{w_{ni}\}_{i=1}^n$, that balances the two terms in the display above, is given as follows: set $k^* = \lfloor B n^{4/d+4} \rfloor$,

3.0 Content Block

$w_{ni}^* = \frac{1}{k^*} \left[1 + \frac{d-2}{d} \frac{k^{*2/d} \{i^{1+2/d} - (i-1)^{1+2/d}\}}{d^2 \{k^{*2/d}\} \{i^{1+2/d} - (i-1)^{1+2/d}\}} \right]$ for $i = 1, 2, \dots, k^*$ and

4.0 Content Block

k-NN outlier The distance to the kth nearest neighbor can also be seen as a local density estimate and thus is also a popular outlier score in anomaly detection. The larger the distance to the k-NN, the lower the local density, the more likely the query point is an outlier. Although quite simple, this outlier model, along with another classic data mining method, local outlier factor, works quite well also in comparison to more recent and more complex approaches, according to a large scale experimental analysis.

5.0 Content Block

Parameter selection The best choice of k depends upon the data; generally, larger values of k reduces effect of the noise on the classification, but make boundaries between classes less distinct. A good k can be selected by various heuristic techniques (see hyperparameter optimization). The special case where the class is predicted to be the class of the closest training sample (i.e. when k = 1) is called the nearest neighbor algorithm. The accuracy of the k-NN algorithm can be severely degraded by the presence of noisy or irrelevant features, or if the feature scales are not consistent with their importance. Much research effort has been put into selecting or scaling features to improve classification. A particularly popular approach is the use of evolutionary algorithms to optimize feature scaling. Another popular approach is to scale features by the mutual information of the training data with the training classes. In binary (two class) classification problems, it is helpful to choose k to be an odd number as this avoids tied votes. One popular way of choosing the empirically optimal k in this setting is via bootstrap method.

6.0 Content Block

In statistics, the k-nearest neighbors algorithm (k-NN) is a non-parametric supervised learning method. It was first developed by Evelyn Fix and Joseph Hodges in 1951, and later expanded by Thomas Cover. In

classification, a new example is assigned a label based on the labels of its k nearest training examples; in regression, the prediction is computed from the values of those neighbors. Most often, it is used for classification, as a k -NN classifier, the output of which is a class membership. An object is classified by a plurality vote of its neighbors, with the object being assigned to the class most common among its k nearest neighbors (k is a positive integer, typically small). If $k = 1$, then the object is simply assigned to the class of that single nearest neighbor. The k -NN algorithm can also be generalized for regression. In k -NN regression, also known as nearest neighbor smoothing, the output is the property value for the object. This value is the average of the values of k nearest neighbors. If $k = 1$, then the output is simply assigned to the value of that single nearest neighbor, also known as nearest neighbor interpolation. For both classification and regression, a useful technique can be to assign weights to the contributions of the neighbors, so that nearer neighbors contribute more to the average than distant ones. For example, a common weighting scheme consists of giving each neighbor a weight of $1/d$, where d is the distance to the neighbor. The input consists of the k closest training examples in a data set. The neighbors are taken from a set of objects for which the class (for k -NN classification) or the object property value (for k -NN regression) is known. This can be thought of as the training set for the algorithm, though no explicit training step is required. A peculiarity (sometimes even a disadvantage) of the k -NN algorithm is its sensitivity to the local structure of the data. In k -NN classification the function is only approximated locally and all computation is deferred until function evaluation. Since this algorithm relies on distance, if the features represent different physical units or come in vastly different scales, then feature-wise normalizing of the training data can greatly improve its accuracy.

7.0 Content Block

Furthest-neighbor variants A variation of the nearest neighbor approach uses the k furthest neighbors instead of the nearest ones. In this setting, neighbors are selected based on maximal dissimilarity rather than similarity. This approach has been proposed in recommender systems as an "inverted neighborhood" model, where the most dissimilar users are identified and their preferences are used inversely to generate recommendations. The goal is typically to increase diversity or novelty in recommended items, since traditional nearest-neighbor methods may favor popular or obvious items. Empirical evaluations have found that furthest-neighbor approaches generally achieve lower predictive accuracy than standard k -nearest neighbor methods, although user-perceived usefulness may be similar or higher in some cases.